Programming Assignment 3: Sensor Reading Variant Record

Data Structures & Algorithms Laboratory Handout (Medium)

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Problem Overview

A monitoring device reports readings from three kinds of sensors: temperature sensors (decimal degrees), door sensors (single character `O` for open or `C` for closed), and pulse counters (large whole number of pulses).

Your program must define a `union` holding all three value types and a `Reading` structure containing a sensor ID, a kind tag character, and the union. After processing `N` readings, output the report and memory footprint comparisons.

Validation Rules:
- Count `N` out of range (0 or below, or above 1000) outputs `INVALID COUNT`.
- Unrecognized kind tags (anything other than `T`, `D`, or `P`) output `INVALID KIND`.

Data Structure Requirements

Variant Record Specifications

- **Union Value:** Holds temperature (float), door state (char), and pulses (long).
- **Reading Structure:** Holds sensor ID (int), kind tag (char), and union.
- **Size Reporting:** Print `sizeof(union Value)`, `sizeof(Reading) with union`, equivalent structure size with all fields active at once, and their difference.

Output Guidelines

Switch on the kind tag to format each report line correctly: temperature to two decimals with a `C` suffix, door state as `OPEN` or `CLOSED`, and pulse count as a whole number.

Sample Test Cases

#	Input (stdin)	Expected Output (stdout)
1	5 11 T 4.50 12 D O 13 P 184320 14 T -2.75 15 D C	SENSOR REPORT Sensor 11 temperature 4.50 C Sensor 12 door OPEN Sensor 13 pulses 184320 Sensor 14 temperature -2.75 C Sensor 15 door CLOSED sizeof(union Value): 8 sizeof(Reading) with union: 16 sizeof(Reading) with all fields: 32 Bytes saved per record: 16
2	1 21 P 9876543210	SENSOR REPORT Sensor 21 pulses 9876543210 sizeof(union Value): 8 sizeof(Reading) with union: 16 sizeof(Reading) with all fields: 32 Bytes saved per record: 16
3	2 51 T 36.60 52 X 5	INVALID KIND

> **Instructor Note:** Ensure proper tag validation before accessing union members to prevent undefined behavior.
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> **— Bavith**